

LogicCircuit can help in creating advanced circuits and sub-circuits as

Using splitters

ping wires and a messy circuit design, you can converge a bunch of the wires into the splitter so that the output draws one clean connection. The number of input lines to the splitter has to be inserted first. By double-clicking the splitter, parameters like bit count, pins, input or output type, and so on

appear in a pop-up. All these parameters determine the splitter's capacity. Any element including constants, input and output pins, random-access memory (RAM) and read-only memory (ROM) supports the bundles.

Inserting memories

The number of columns are the square of the value of the address bit entered, therefore these change dynamically as per the address bit-width entered. Data cells store data in hexadecimal format. You can also load data from a pre-stored binary file.

The RAM circuit comes with three input pins and one output pin. Among the three inputs, the first pin accepts the address of the cell that will store or read data. The second pin accepts the data from the circuit at given time intervals, which will be stored in the memory cell. The third pin is the write-signal acceptor of the RAM. RAM functional properties can be edited by double-clicking the element.

RAM editor has similar parameters as ROM, like address bit-width, data bit-width and data cells. There are two more properties. 'Write on' property, available as a dropdown, determines when the RAM will write data into its memory. 'On Start Fill' sets the RAM characteristics when the circuit will be powered up and the RAM will start running. Select suitable options to determine the complete functional nature of your RAM.

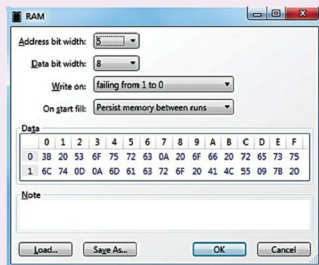


Fig. 3: RAM editor

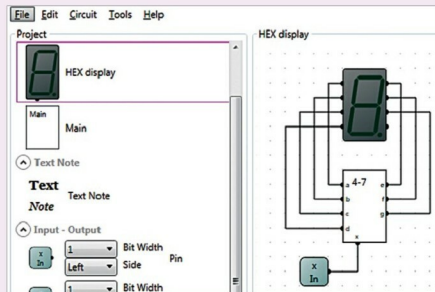


Fig. 4: Creating a visual display